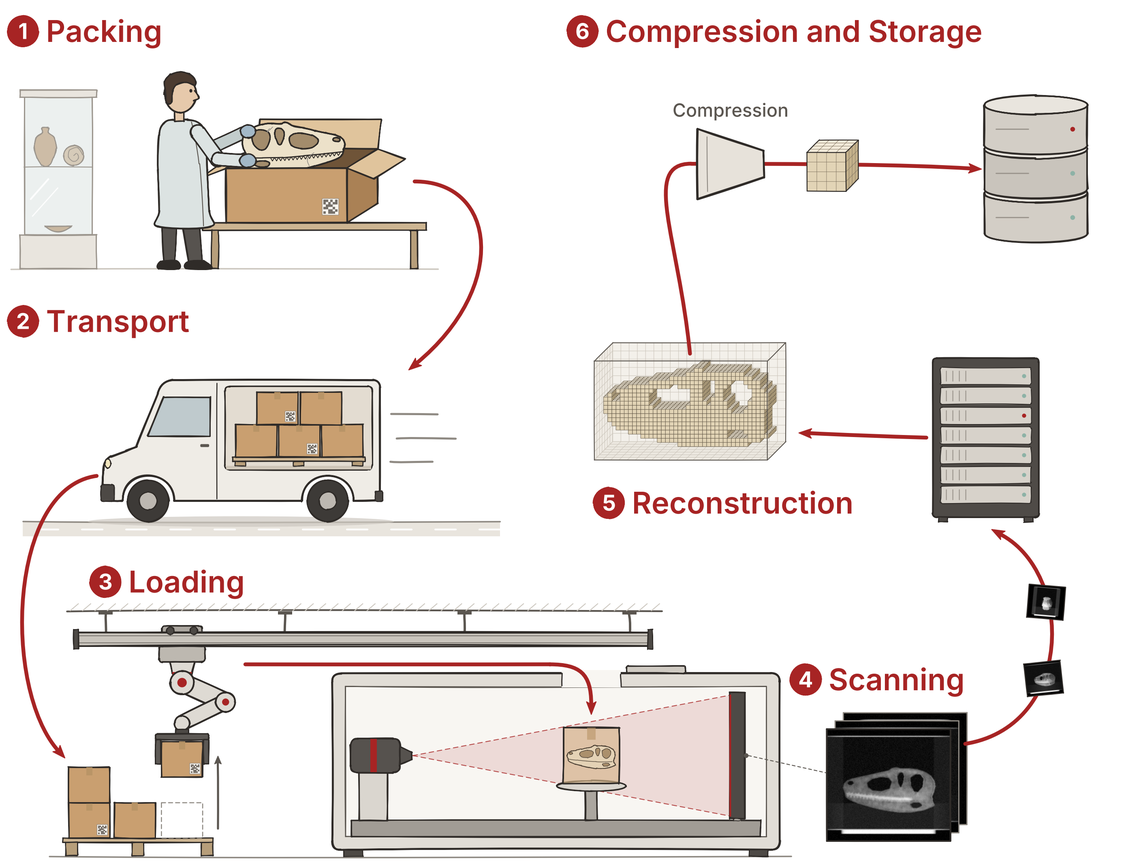
Compression Strategies for High-Throughput X-Ray CT of Natural History Collections

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1 Introduction

The *NaturArv3D* pilot project aims to **digitise the vertebrate collection** at Natural History Museum Denmark (NHMD) in collaboration with the 3D Imaging Centre, Technical University of Denmark (3DIM).

As necessitated by the scale of the collection, a custom high-throughput X-ray µCT scanner with a robotic sample handling system is being developed at 3DIM with a designed capacity of **500 samples per week**.



The proposed *NaturArv3D* pipeline for high-throughput X-ray µCT of museum artefacts.

2 Background

- Up to $26\,000$ samples per year
- sub-$10\,\mu\mathrm{m}$ resolution at 3072^3 to 4800×5606^2 voxels per sample
  - $58 - 302\ \mathrm{GB/sample}$ for 16-bit data; $116 - 603$ for 32-bit
- 1 PB of raw data per year in optimistic scenario†

Assuming a storage cost of $600\ \mathrm{DKK/(TB\ year)}$, the amortised **cost of storing 1 year's worth of data is 2.6 million DKK**.‡

This calls for compression, which must generally be lossy to exceed a compression ratio (CR) of $2{:}1$ [1]. The archival format must further be **future-proof, scalable, and supported by the scientific community**.

3 Methods

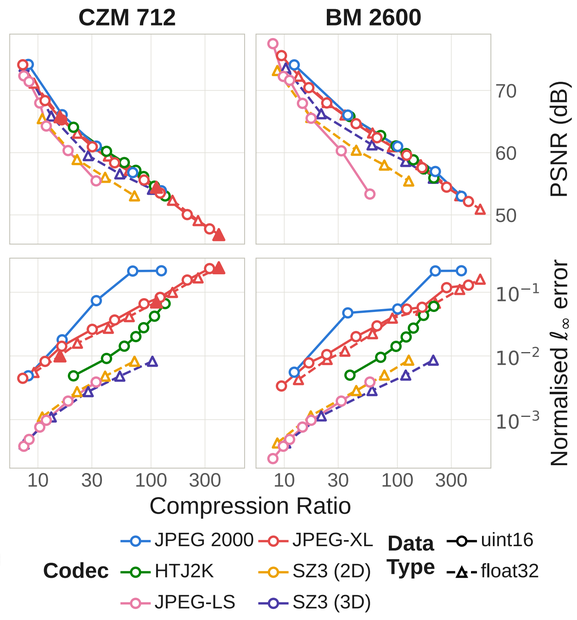
Three representative specimens, each as both a 16-bit unsigned-integer and a 32-bit floating-point volume

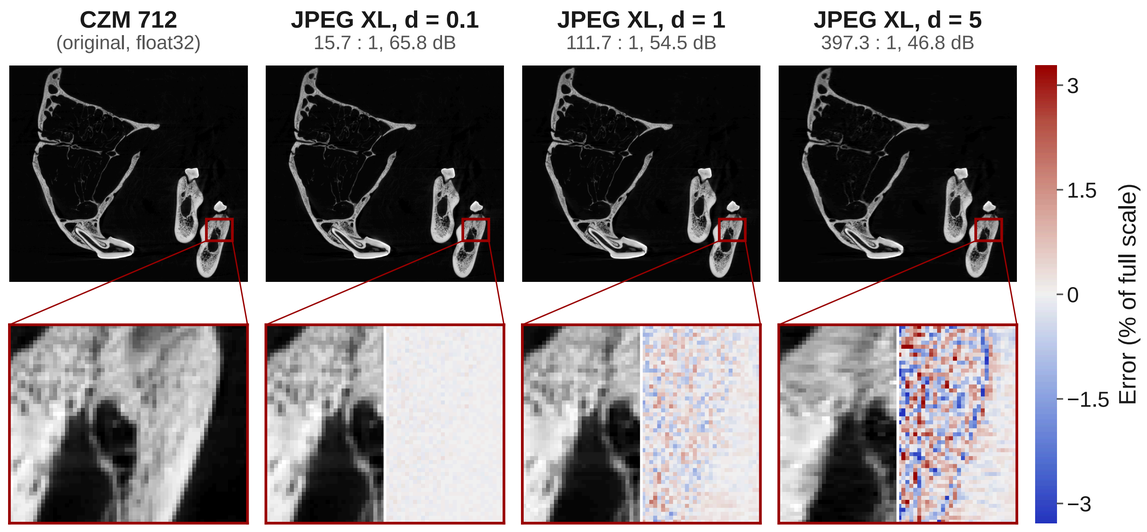
- Downscaled to 1000^3 voxels for computational feasibility
- 106 combinations of container format and compression configurations
  - DICOM, **OME-Zarr**
  - **JPEG-LS**, **JPEG 2000**, **HTJ2K**, **JPEG XL**, LERC, ZFP, **SZ3**, SPERR, Pcodec
- Uncompressed 16-bit TIFF stacks provide baseline for comparison
- Fidelity via full-reference measures: ℓ_∞; **PSNR**; 3D SSIM, MS-SSIM [2]

† 16-bit, 50% crop, 50% facility utilisation rate, 70% scans with small detector.
‡ Amortisation of storage over 5 years assuming fixed price.

4 Results

- **High CR at modest quality loss**
- Transform coders (JPEG XL/2000, HTJ2K) superior in PSNR
- Error-bounded codecs (JPEG-LS, SZ3) superior in ℓ_∞
- Only SZ3 supports 3D chunks
  - Moderate improvement over 2D
- Float representation has no cost when compressed
- *Similar performance across specimens*
- *(MS-)SSIM follow PSNR (not shown)*





For the JPEG XL codec (`effort=7`) we observe:

- Errors *across* compression planes show no significant artefacts due to slicewise compression
- Perceptible loss in quality at higher compression levels ($d = 5$)
- Acceptable fidelity at $d \leq 1$, still offers impressive performance
  - $\mathrm{CR} = 111.7$ equivalent to $\mathbf{99.1\%}$ **space saving**

5 Findings and Conclusion

DICOM is currently the only *widely* supported container with lossy compression (JPEG 2000/LS/XL).

Modern formats like OME-Zarr are seeing rapid adoption and improve scalability through **chunking, streaming, and multi-resolution hierarchies**.

As an example, JPEG XL with $d = 0.5$ offers a space saving of 98.3% ± 0.4% ($N = 6, \mathrm{CR} = 62.4 \pm 16.3$) without a significant reduction in quality.

For Natural History Museum Denmark, this conservatively corresponds to an estimated **annual cost saving of 2.5 million DKK** from the *NaturArv3D* project alone.

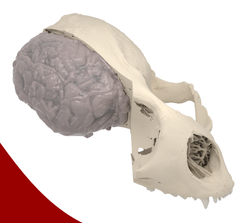
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6 References

[1] F. Marone, J. Vogel, and M. Stampanoni, "Impact of Lossy Compression of X-ray Projections onto Reconstructed Tomographic Slices," *Journal of Synchrotron Radiation*, vol. 27, no. 5, pp. 1326–1338, Sep. 2020, doi: 10.1107/S1600577520007353.

[2] Z. Wang, E. Simoncelli, and A. Bovik, "Multiscale Structural Similarity for Image Quality Assessment," in *The Thrity-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*. Nov. 2003, p. 1398–1402 Vol.2. doi: 10.1109/ACSSC.2003.1292216.

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